

Ayrton Porto

AI Engineer · Formal Methods · LLM Systems

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Summary

Mathematician and AI engineer building LLM-powered systems for automated reasoning and formal verification. Strong background in mathematical logic, hands-on experience with large language models, multi-agent pipelines, and proof assistants (Lean 4, Coq). Built production data pipelines and a web application with FastAPI. Seeking roles in AI engineering, research engineering, or formal methods.

Technical Skills

Languages	Python, C++, SQL, LaTeX
ML / AI	LLM systems, multi-agent systems, API integration, PyTorch
Web	FastAPI, Streamlit, SSE streaming, REST APIs
Data	Pandas, statistical analysis, data pipelines, web scraping
Formal	Lean 4, Coq / Rocq, TLA+, Dafny, Z3
Tools	Git, GitHub

Languages

Spanish (native) · English (professional proficiency)

Selected Projects

AViD Journal — LLM Autoformalization Pipeline 2025–Present
Independent Research Project — Deployed, in active development (avidjournal.com.ar)

- End-to-end system formalizing mathematical papers in Lean 4 via the Claude API.
- Designed a novelty-detection pipeline that evaluates formalized theorems against Mathlib along three dimensions: existence, non-triviality, and structural proof-distance.
- Modular Python backend with FastAPI + SSE streaming for real-time pipeline progress.
- github.com/ayrtonporto/avid-journal

ManaProof — Data-Driven Commander Deck Builder 2024–Present

- Built a data pipeline integrating EDHREC data and large-scale card statistics to analyze card popularity, synergies, and deck composition for automated Magic: The Gathering Commander deck construction.
- Designed a data-driven deck optimization pipeline combining statistical analysis, card-synergy modeling, and automated deck construction.
- Built and sold 250+ automated Commander decklists through the resulting workflow.
- github.com/ayrtonporto/manaproof

Experience

Apart Research — Fellow Researcher 2026–Present
Secure Program Synthesis Fellowship · International collaboration

- Developing **BabelBench**, a cross-paradigm proof-translation benchmark measuring whether LLMs preserve logical meaning when translating theorems across formal systems (Lean 4, Coq, TLA+, Dafny).
- Designed fidelity metrics checked by real proof verifiers and the Z3 solver — no LLM-based evalua-

tion.

— Collaborating with mentors and co-researchers on benchmark design and evaluation methodology.

ManaProof — Founder & AI Engineer 2024–Present

— Founded and operate a data-driven Magic: The Gathering deck-building business.

— Built a data pipeline integrating EDHREC data and large-scale card statistics for automated Commander deck optimization.

— Designed statistical models for card-synergy analysis and deck composition.

Teaching Assistant — **UNICEN** 2023–Present

Graduate TA (2026–) · Undergraduate TA (2023–2025)

Courses: Discrete Mathematics, Linear Algebra, Calculus I.

Education

UNICEN — Licenciatura en Ciencias Matemáticas 2020–2025

GPA: 9.41/10.00. Five-year degree, equivalent to B.S. + M.S.

Thesis: Priestley duality for distributive lattices.

Awards

Apart Research Fellow 2026

Santander Scholarship — Academic Merit 2024

CIN Scholarship — Consejo Interuniversitario Nacional 2023–2025

IMPA Summer Program — Rio de Janeiro, Brazil 2023

Honorable Mention — Argentine Mathematical Olympiad (OMA) 2017, 2019